

UNIVERSIDAD NACIONAL MAYOR DE SAN MARCOS
FACULTAD DE INGENIERÍA DE SISTEMAS E INFORMÁTICA
DOCTORAL PROGRAM IN DEEP TECH, ARTIFICIAL INTELLIGENCE & EMERGING
TECHNOLOGIES



RESEARCH METHODS AND SCIENTIFIC INTEGRITY IN AI AND ADVANCED
TECHNOLOGIES

Deliverable:

Method Fit Matrix

Professor:

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Student:

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Scoped Research Question: How can an Explainable Artificial Intelligence approach be developed and rigorously evaluated to support the early identification of the risk of school interruption among secondary education students in Peru, considering predictive performance, explanation faithfulness and stability, fairness, and responsible educational use?

CRITERION	Weight	Candidate Design Research	A: Science	Candidate Experimental / Evaluation	B: Quasi- Empirical ML	Candidate Action Research	C: Design	Your one-line justification
E • Epistemological fit	30	5/5		4/5		5/5		DSR and ADR strongly fit the pragmatic, solution-oriented position; empirical ML evaluation also provides valid evidence, but does not fully govern artifact design and refinement.
D • Data availability	20	4/5		4/5		3/5		DSR and empirical ML evaluation can initially use available educational datasets, while ADR additionally requires sustained access to schools, teachers, and other educational stakeholders.
F • Feasibility	15	4/5		5/5		3/5		Empirical ML evaluation is the easiest to execute computationally; DSR remains feasible with a bounded artifact, while ADR requires greater institutional coordination, field access, ethics approval, and iterative participation.
C • Contribution type	20	5/5		3/5		5/5		DSR and ADR can generate an evaluated artifact and transferable design knowledge; empirical ML evaluation mainly produces comparative evidence about model performance.
V • Venue fit	15	5/5		5/5		4/5		DSR and empirical ML studies fit AI-in-Education, XAI, EdTech, and information-systems venues; ADR is also publishable but is more naturally aligned with participatory design, HCI, and information-systems venues.
GATE: $E \leq 2$ eliminates	—	SURVIVES		SURVIVES		SURVIVES		All three candidates are epistemologically defensible; therefore none is eliminated by the gate.
Raw total	—	23/25		21/25		20/25		The raw total is informative but does not determine the methodological decision.
Weighted total	100	93/100		82/100		83/100		DSR obtains the strongest overall fit with the intended doctoral contribution.